An Anchor-Free Method Based on Feature Balancing and Refinement Network for Multiscale Ship Detection in SAR Images

Jiamei Fu, *Graduate Student Member, IEEE* , Xian Sun, *Senior Member, IEEE* , Zhirui Wang, *Member, IEEE*, and Kun Fu, *Member, IEEE*

***Abstract*— Recently, deep-learning methods have been successfully applied to the ship detection in the synthetic aperture radar (SAR) images. It is still a great challenge to detect multiscale SAR ships due to the broad diversity of the scales and the strong interference of the inshore background. Most prevalent approaches are based on the anchor mechanism that uses the predefined anchors to search the possible regions containing objects. However, the anchor settings have a great impact on their detection performance as well as the generalization ability. Furthermore, considering the sparsity of the ships, most anchors are redundant and will lead to the computation increase. In this article, a novel detection method named feature balancing and refinement network (FBR-Net) is proposed. First, our method eliminates the effect of anchors by adopting a general anchor-free strategy that directly learns the encoded bounding boxes. Second, we leverage the proposed attention-guided balanced pyramid to balance semantically the multiple features across different levels. It can help the detector learn more information about the small-scale ships in complex scenes. Third, considering the SAR imaging mechanism, the interference near the ship boundary with the similar scattering power probably affects the localization accuracy because of feature misalignment. To tackle the localization issue, a feature-refinement module is proposed to refine the object features and guide the semantic enhancement. Finally, extensive experiments are conducted to show the effectiveness of our FBR-Net compared with the general anchor-free baseline. The detection results on the SAR ship detection dataset (SSDD) and AIR-SARShip-1.0 dataset illustrate that our method achieves the state-of-the-art performance.**

***Index Terms*— Attention-guided balanced pyramid, feature balancing and refinement network (FBR-Net), feature refinement, ship detection, synthetic aperture radar (SAR).**

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Jiamei Fu, Xian Sun, and Kun Fu are with Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100190, China, and also with the Key Laboratory of Network Information System Technology (NIST), Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China (e-mail: fujiamei14@mails.ucas.ac.cn; sunxian@mail.ie.ac.cn; fukun@mail.ie.ac.cn).

Zhirui Wang is with Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, and also with the Key Laboratory of Network Information System Technology (NIST), Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China (e-mail: zhirui1990@126.com).

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I. Introduction

SYNTHETIC aperture radar (SAR) is a kind of active microwave imaging sensor with the capability of providing high-resolution images in all-day and all-weather conditions. It has been widely used in many fields [1]–[5]. Among the SAR applications, ship detection plays an important role in maritime management and monitoring. In recent years, with the development of airborne SAR and spaceborne SAR, a lot of studies [6]–[12] have been conducted in ship detection. Multiscale SAR ship detection in complex scenes is one of the main tasks, which is still a great challenge.

The traditional approaches for SAR object detection mainly include the methods based on the contrast information [13]–[15], the geometric and texture features [16], [17], as well as the statistical analysis [18], [19]. Among the existing detection algorithms, the constant false-alarm rate (CFAR) method [20] is one of the most commonly used techniques. The CFAR method computes the adaptive threshold based on the given false-alarm rate and the statistical distributions of the background clutter, and then distinguishes the object pixels from the background by comparing the pixel intensity with the calculated threshold. The performance of this method largely depends on the statistical modeling of the sea clutter and the parameter estimation of the selected models. Mainly focusing on these two aspects, a lot of improved methods have been proposed. Considering that the clutter is nonuniform, various models, such as the symmetric alpha-stable distribution [21] and the generalized gamma distribution (GΓD) [22], are based on the heterogeneous distributions to fit the variable sea conditions. Nevertheless, as the model complexity increases, the parameter estimation becomes difficult and time-consuming. Gao *et al.* [23] considered the practical application, trying to strike a good balance between the estimation precision and the speed. In general, the CFAR methods achieve good performances when the scenes are relatively simple. However, it is difficult to deal with the small ships and the inshore complex scenes. Inappropriate modeling of the inshore areas results in many inland false alarms as well as the performance degradation. In order to improve the robustness of the ship detectors, a series of preprocess including the speckle filtering and the land masking is usually used. However, the coastline extraction result has a direct impact on the performance of the ensuing detection for ships near the

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coast, which also prevents the whole pipeline from being end to end.

Benefitting from the development of deep learning and GPU computing power, deep convolutional neural networks (CNNs) have made remarkable breakthroughs in object detection. Compared with the traditional methods using manually designed features, deep neural networks can learn the feature representations automatically from the given data. Modern CNN-based detectors can be divided into two categories: the anchor-based methods and the anchor-free methods. Inspired by the idea of object proposals, the anchor-based detectors such as Faster R-CNN [24], Cascade R-CNN [25], SSD [26], and RetinaNet [27] adopt the anchor boxes as common references to search the possible regions containing object of interests. These anchor boxes are manually designed according to the statistical characteristics of the data set in order to deal with the scale and ratio variation. In comparison, the anchor-free methods explore an alternative direction by eliminating the predefined anchor boxes and become proposal-free. For instance, CornerNet [28] and ExtremeNet [29] directly detect several key points of the bounding boxes. FCOS [30] and FoveaBox [31] tackle the detection problem in a per-pixel prediction manner. The anchor-based methods play a dominant part in object detection, while the anchor-free methods are still under development. Even so, the anchor-free methods have shown a great potential to surpass the anchor-based counterparts and achieve a better speed–accuracy tradeoff.

On account of the powerful feature extraction and representation ability, the CNN-based methods have been introduced to detect objects in the remote sensing images [32]–[34]. As for the SAR object detector, deep CNNs were initially applied in certain parts of the conventional methods, such as generating the proposals for the guard windows of the CFAR [35], separating the sea from the land [36], and identifying different objects [37]. Recently, various end-to-end methods are proposed to enhance the efficiency and accuracy. Kang *et al.* [38] added contextual features into the corresponding regions of interest so that the environmental information can help to rule out the false alarms. Jiao *et al.* [39] fused the features with different resolutions through the dense connection to generate the multiscale features with more spatial and semantic information. In order to obtain the salient features and suppress the clutter, some studies adopt the attention mechanism. Cui *et al.* [40] integrated the feature pyramid with the convolutional block-attention module. Chen *et al.* [41] proposed an attention module with two parallel branches. Du *et al.* [42] introduced the saliency information into the network so that the detector can focus more on the object areas. Zhao *et al.* [43] proposed a dilated attention block to enhance the feature extraction ability of the detector. Different from the methods with horizontal bounding boxes, Liu *et al.* [44], An *et al.* [45], and Chen *et al.* [46] used the oriented bounding boxes, which perform better for the densely arranged objects. The improvements above are made on the basis of the anchor-based detectors. Although these methods have improved the performance to a certain degree, there are still some drawbacks. First, the anchor boxes introduce extra hyperparameters. They need careful manual tuning, because the detection performance is sensitive to the anchor settings. Second, one fixed set of anchor boxes is not always suitable for other data sets with different characteristics, especially when the objects have large-scale variation. Third, in order to maintain a high recall, a large number of anchor boxes are densely sampled over the input image at the cost of a heavy computation load. Nevertheless, most of them are redundant due to the sparsity of the ships in the SAR images.

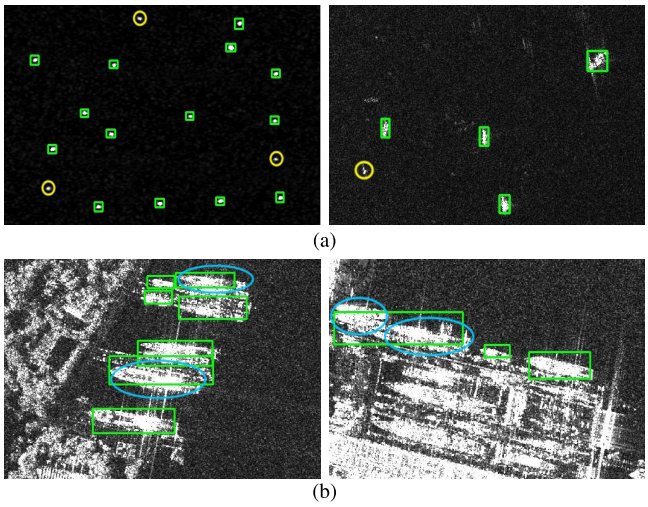


Fig. 1. Some detection results of the general anchor-free detector. The green rectangles are the detection results. The yellow and blue circles represent the missing ships and the ships with the inaccurate bounding box, respectively. (a) Problem of the missing ships. (b) Problem of inaccurate localization.

The aforementioned effect of anchor boxes inspires us to adopt an anchor-free method, but the recent general anchor-free detectors [30], [31] well used in the optical images do not perform well for the multiscale ship detection in the SAR images. One of the important problems is that the small-scale ships are likely to be ignored, as shown in Fig. 1(a). Small objects are usually detected using the high-resolution features in low levels, while the low-level features are semantically weak in comparison with the high-level features. The inconsistency across different levels makes the detectors pay more attention on the salient objects with sufficient semantic information and miss the small objects. Pyramidal feature representations are one of the basic solutions to address this imbalance issue. The feature pyramid network (FPN) [47] pioneeringly combines the adjacent levels using lateral connections along the top-down pathway. Based on the FPN, PANet [48] adds an extra bottom-up pathway to enhance further the low-level features. The Libra R-CNN [49] integrates the feature maps from all FPN levels, which can be refined to generate more balanced semantic features. Unlike the top-down structure, the scale-transferrable detection network (STDN) [50] proposes a scale-transfer module to make full use of the backbone layers. The parallel feature pyramid network (PFPNet) [51] widens the network by proposing a parallel structure with an additional transformation operation, which can yield features with the similar semantic information. Azimi *et al.* [52] proposesd a novel joint image cascade and FPN to realize the multiscale feature extraction. The NAS-FPN [53] automatically designs a more powerful feature fusion network through the neural

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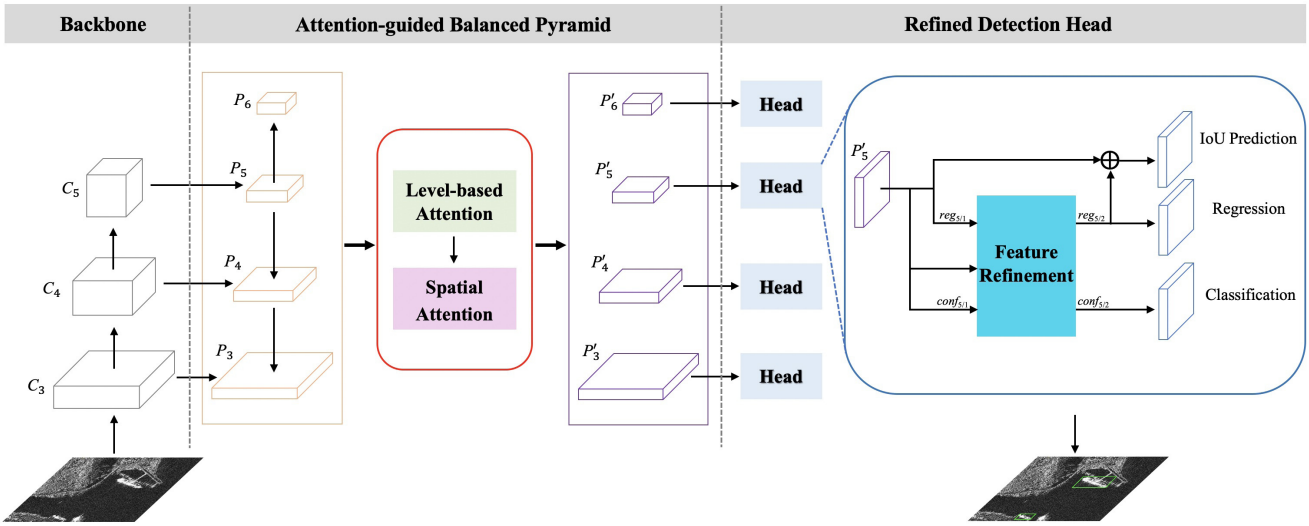


Fig. 2. Overall architecture of FBR-Net. It consists of the backbone network, the ABP, and the refined detection head.

architecture search algorithm. These methods have proved that multiscale feature balancing plays a critical role in detecting the small-scale objects. Another important problem of directly applying the general anchor-free networks to SAR ship detection is the inaccurate localization of some ships, as shown in Fig. 1(b). In the SAR images, the ships generally present different scattering power distributions from the background. However, some local areas of ships have the similar scattering mechanism with certain parts of the surrounding area. For example, both the ship deck and the sea surface have a relatively low scattering intensity, and the metal parts of the ships have the similar strong scattering phenomenon with some facilities in the harbors. These factors will lead to the results that some predicted bounding boxes cannot cover the intact object accurately. We identify that the semantic representations of the local areas close to the ship boundary are susceptible to the adjacent interference with the similar scattering power, and the corresponding features with the low semantic information are not well aligned with the object. This misalignment further results in an inaccurate bounding box.

According to the above analysis, we propose an anchor-free method for detecting the multiscale SAR ships in complex scenes called feature balancing and refinement network (FBR-Net). More specifically, we improve the general anchor-free detector by two novel structures with the aim to tackle the two issues of ship detection in the SAR images, respectively. The main contributions of our work can be summarized as follows.

1) Different from the most anchor-based approaches, we present an anchor-free detection framework for the multiscale SAR ship detection, which eliminates the effect of anchors by directly learning the encoded bounding boxes.
2) To focus more on the small-scale ships, an attention-guided balanced pyramid (ABP) is proposed to balance semantically the features across different pyramidal levels. In this procedure, the salient features in complex scenes can be extracted adaptively and efficiently.
3) Considering that the features close to the ship boundary are susceptible to the adjacent interference with a similar scattering power in the SAR images, we design a feature-refinement (FR) module to refine the object features and mitigate the misalignment issue, which increases the localization accuracy to a certain degree.

The experiments on the SAR ship detection dataset (SSDD) and AIR-SARShip-1.0 data sets show that the proposed method achieves a better detection performance than the prevalent detectors, which demonstrates the effectiveness of our model.

The rest of this article is organized as follows. Section II illustrates the proposed method in detail. In Section III, the experimental results and the corresponding analysis are provided. Finally, some conclusions are drawn in Section IV.

II. Methodology

The overall architecture of our proposed network is shown in Fig. 2. It can be divided into three components: the backbone network for feature extraction, the ABP for further feature representation, and the refined detection head for final prediction. In this part, we first introduce the general anchor-free method, which is adopted as the baseline in our work. Next, the ABP structure including two subnets is described in detail. Then, we present our detection head and illustrate the design of the proposed FR module. The loss function is given at the end of this section.

A. General Anchor-Free Baseline

Recent anchor-free detectors [30], [31] are based on a general structure, in which the detection head with two task-specific subnets is attached to the different levels in the feature pyramid. The feature maps at pyramid level l are denoted as $P_l \in \mathbb{R}^{(W/s_l)\times(H/s_l)\times C}$ with $l = 3, 4, 5, 6$, where $W \times H$ is the size of the input image and C is the channel dimension of P_l (C is set as 256), s_l is the corresponding downsampling ratio to the input image, and $s_l = 2^l$. As shown in Fig. 3, both branches in the detection head have

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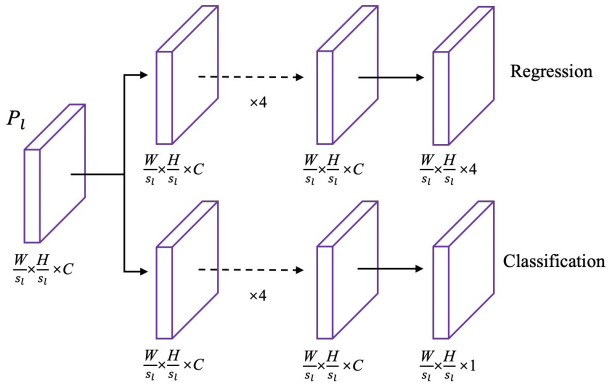


Fig. 3. Structure of general detection head.

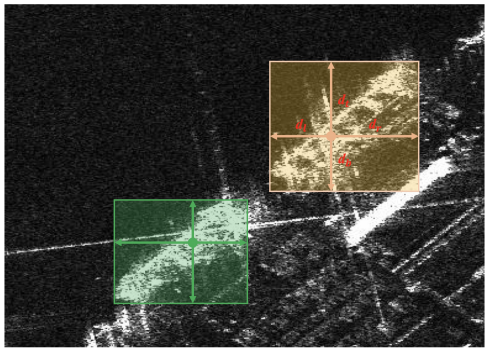


Fig. 4. Encoded location of the bounding box.

five 3 × 3 convolutional layers. One branch performs binary classification at each pixel, predicting the possibility of object existence, and the other one predicts the encoded bounding boxes.

We define the pixel point at location (x, y) on P_l as p^l_{xy} ($x = 0, 1, \ldots, W/s_l - 1$ and $y = 0, 1, \ldots, H/s_l - 1$). For each p^l_{xy}, it can be mapped back onto the input image location (x', y'), where $x' = s_l(x+0.5)$ and $y' = s_l(y+0.5)$. Following a similar strategy as in [30], the predicted bounding box is encoded by a 4-D vector $\boldsymbol{v} = (d_l, d_t, d_r, d_b)$, depicting the relative distance from the location (x', y') to the four boundaries, as shown in Fig. 4. If location (x', y') is inside a ground-truth box $B = (x_0, y_0, x_1, y_1)$, the pixel point p^l_{xy} is regarded as a positive sample and the class label $\hat{c} = 1$, where (x_0, y_0) and (x_1, y_1) are the coordinates of the top-left and bottom-right corners, respectively. Otherwise, it is a negative sample and $\hat{c} = 0$. For a positive pixel point, its regression target $\hat{\boldsymbol{v}} = (\hat{d}_l, \hat{d}_t, \hat{d}_r, \hat{d}_b)$ can be calculated as

$$\begin{aligned} \hat{d}_l &= x' - x_0, \quad \hat{d}_t = y' - y_0 \\ \hat{d}_r &= x_1 - x', \quad \hat{d}_b = y_1 - y'. \end{aligned} \tag{1}$$

Ships in different scales are assigned to different pyramid levels. Specifically, $[m_{l-l}, m_l]$ is the scale range that level l is responsible for regression. If the maximum value among $\hat{d}_l, \hat{d}_t, \hat{d}_r$, and $\hat{d}_b$ is not in this range, then the corresponding pixel point is considered as a negative sample. In this article, m_2, m_3, m_4, m_5, and m_6 are set as 0, 64, 128, 256, and ∞, respectively. If a pixel point is assigned to more than one object, the bounding box with a minimal area is selected as the ground truth. To keep the detector efficient, the detection heads between the different levels are shared, and an exponential function with a learnable parameter is used to adjust different feature levels to the multiscale detection.

B. Attention-Guided Balanced Pyramid

In the deep CNNs, the high-level feature maps with the more semantic and less spatial information usually have larger receptive fields, which are suitable for the large-scale ship detection. By contrast, the feature maps in low levels with relatively more spatial information and smaller receptive fields are suitable to detect small-scale ships. This imbalance across different levels makes the detectors focus more on the salient objects instead of the small objects. Therefore, how to use the features of different levels to generate appropriate pyramidal representations is crucial to the multiscale ship-detection performance. In [49], an effective method named balanced feature pyramid (BFP) is proposed to learn the residual features by integrating and refining all the FPN levels. Considering that each level contributes to the balanced semantic features unequally, different from the direct rescaling and averaging operation in [49], our basic idea is to obtain the balanced feature maps adaptively. The proposed ABP structure contains two subnets: the level-based attention (LA) and the spatial attention (SA). The LA module is designed to learn adaptively the level-based weights of fusion for the resized feature maps from all the FPN levels. The SA module aims to refine further the balanced maps and focus on the salient regions containing ships.

1) Level-Based Attention Module: We use ResNet-50 [54] as the backbone, and it is divided into five stages according to the size of the feature maps in different layers. In the bottom-up pathway, the outputs of the last three stages are defined as $\{C_3, C_4, C_5\}$. As shown in Fig. 2, $\{P_3, P_4, P_5\}$ in the top-down pathway are produced by the corresponding bottom-up layer through the lateral connection, which is the same as in [47]. This process is summarized as

$$I_l = \begin{cases} \text{Conv}_{1\times1}(C_5), & l = 5 \\ \text{Upsample}(I_{l+1}) + \text{Conv}_{1\times1}(C_l), & l = 3, 4 \end{cases} \tag{2}$$

$$P_l = \text{Conv}_{3\times3}(I_l), \quad l = 3, 4, 5 \tag{3}$$

where $I_l \in \mathbb{R}^{(W/s_l)\times(H/s_l)\times C}$ represents the merged features, $\text{Conv}_{1\times1}$ and $\text{Conv}_{3\times3}$ denote the 1×1 and 3×3 convolutional layer, and Upsample is the nearest neighbor upsampling with the factor of 2. Moreover, a 3×3 convolutional layer with the stride of 2 is attached on P_5 to generate P_6. We choose P_5 as an intermediate level and resize other feature maps to the same size as P_5 with the max pooling and interpolation strategies, respectively. The feature vector at location (i, j) on the resized P_l is denoted as $\boldsymbol{a}^l_{ij}$ ($i = 0, 1, \ldots, W/s_5 - 1$ and $j = 0, 1, \ldots, H/s_5 - 1$), and the corresponding vector $\boldsymbol{b}^l_{ij}$ on the output-balanced feature maps $F_{\text{L}} \in \mathbb{R}^{(W/s_5)\times(H/s_5)\times C}$ is generated as follows:

$$\boldsymbol{b}^l_{ij} = \sum_{l=3}^{6} \lambda^l_{ij} \cdot \boldsymbol{a}^l_{ij} \tag{4}$$

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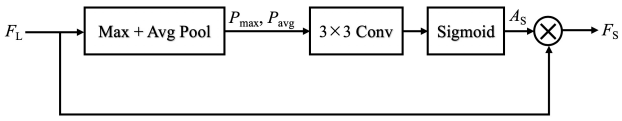


Fig. 5. Structure of SA module.

where λ_{ij}^l refers to the normalized scalar weight at location (i, j) for level l, which can be adaptively learned by the network. To be specific, we first integrate all the resized FPN levels by concatenation. Then, the 1×1 and 3×3 convolutional layers are applied to reduce the channel dimension and produce the weight map $W_l \in \mathbb{R}^{(W/s_5)\times(H/s_5)\times 1}$ for each level. Parameter λ_{ij}^l is computed using the softmax function

$$\lambda_{ij}^l = \frac{\exp\left(\omega_{ij}^l\right)}{\sum_{k=3}^{6}\exp\left(\omega_{ij}^k\right)} \tag{5}$$

where ω_{ij}^l is the corresponding scalar weight on W_l before the normalization. The LA module integrates both semantic and spatial information from all the FPN levels. During this process, the significant features for the multiscale ship detection can be selected adaptively.

2) Spatial Attention Module: To refine further the balanced feature maps F_L obtained by the LA module, as shown in Fig. 5, the SA module first generates the features $P_{max}, P_{avg} \in \mathbb{R}^{(W/s_5)\times(H/s_5)\times 1}$ by using the max and the average pooling along the channel axis, respectively, and aggregates the results through a concatenation operation. Then, one 3×3 convolutional layer and the sigmoid function are applied to obtain the SA feature map $A_S \in \mathbb{R}^{(W/s_5)\times(H/s_5)\times 1}$. This process is computed as follows:

$$\begin{aligned} A_S(F_L) &= \sigma\left(\mathrm{Conv}_{3\times3}([\mathrm{MaxPool}(F_L); \mathrm{AvgPool}(F_L)])\right) \\ &= \sigma\left(\mathrm{Conv}_{3\times3}([P_{max}; P_{avg}])\right) \end{aligned} \tag{6}$$

where MaxPool and AvgPool represent the max pooling and average pooling, and σ is the sigmoid function. Finally, the input feature maps F_L are multiplied by the attention map as follows:

$$F_S = A_S(F_L) \otimes F_L \tag{7}$$

where $\otimes$ represents the elementwise multiplication. After that, $F_S \in \mathbb{R}^{(W/s_5)\times(H/s_5)\times C}$ is resized to the same size as P_3, P_4, and P_6, respectively. Then, it can be used to strengthen the original FPN levels through elementwise addition. The outputs $\{P_3', P_4', P_5', P_6'\}$ are used for the following multiscale ship detection.

C. Refined Detection Head

As for the inaccurate localization problem, recent advances [55]–[57] mainly focus on the approaches to define the confidence of the predicted bounding boxes. Considering the scattering mechanism, we identify that the feature misalignment is another reason responsible for the localization issue in the SAR images. To improve the localization accuracy, we refine the general detection head with two kinds of components.

1) IoU-Prediction Branch: In the standard process of nonmaximum suppression (NMS), all the detection results are ranked according to their classification confidence while ignoring the localization accuracy. In other words, a predicted box with low classification confidence but high intersection-over-union (IoU) will be suppressed by the bounding box with high classification confidence but low IoU. In [30], a single-layer centerness branch parallel to the classification branch is proposed to reduce the weights of the locations close to the object boundary. During the inference, the confidence is multiplied by the corresponding centerness score. Although it suppresses some low-quality predictions produced by the pixel points far from the object center, there is still a problem of low correlation between the classification and the localization.

Inspired by Jiang *et al.* [55] and Wu *et al.* [57], we improve the baseline detection head with an additional IoU-prediction branch in parallel to the regression subnet, as shown in Fig. 2, which aims to predict the IoU between the ground truth and the regressed bounding box. The IoU score can be regarded as a kind of localization confidence, and it is helpful for reducing the weights of inaccurate bounding boxes adaptively. The IoU-prediction subnet applies one 3×3 convolutional layer to the pyramidal feature maps, and the results are then merged with the fourth regression layer using the elementwise addition. We also consider some other choices related to the design of the IoU branch, such as directly introducing a single-layer branch like [30] or designing another independent task-specific subnet the same as the regression and classification branch. In comparison, our design achieves a higher detection accuracy and is relatively parameter-efficient. At inference, the final detection confidence of each pixel point p_{xy}^l is calculated by multiplying the predicted confidence score and the corresponding IoU-prediction score.

2) Feature-Refinement Module: Unlike the objects in the natural images with the more detailed texture information, SAR objects are usually reflected by the different distributions of the scattering intensity from the background. It is relatively easy for the detector to determine the rough location of the objects. However, for certain local areas on the ships, they present a similar scattering mechanism with some parts of the surrounding background, which may have an impact on the precise object localization. Take the ship in Fig. 6(c) for instance, the ship deck has a similar low scattering intensity with the sea clutter, and the foremost part of the ship is not included by the predicted bounding box. We argue that this problem lies in the weak semantic representations of the object areas close to the boundary, since their receptive fields contain much background information and they are susceptible to the adjacent interference with the similar scattering power. As shown in Fig. 6(b) and (f), the ship center indicates higher scores in the heatmaps, while the boundary areas are underrepresented and show a similar score with the surrounding context. The corresponding features with the weak semantic information are not well aligned with the object, further leading to the inaccurate localization, as shown in Fig. 6(c) and (g).

In order to address the above misalignment issue, we propose an FR module. As shown in Fig. 2, the FR module splits all the convolutional layers of the original head into two parts.

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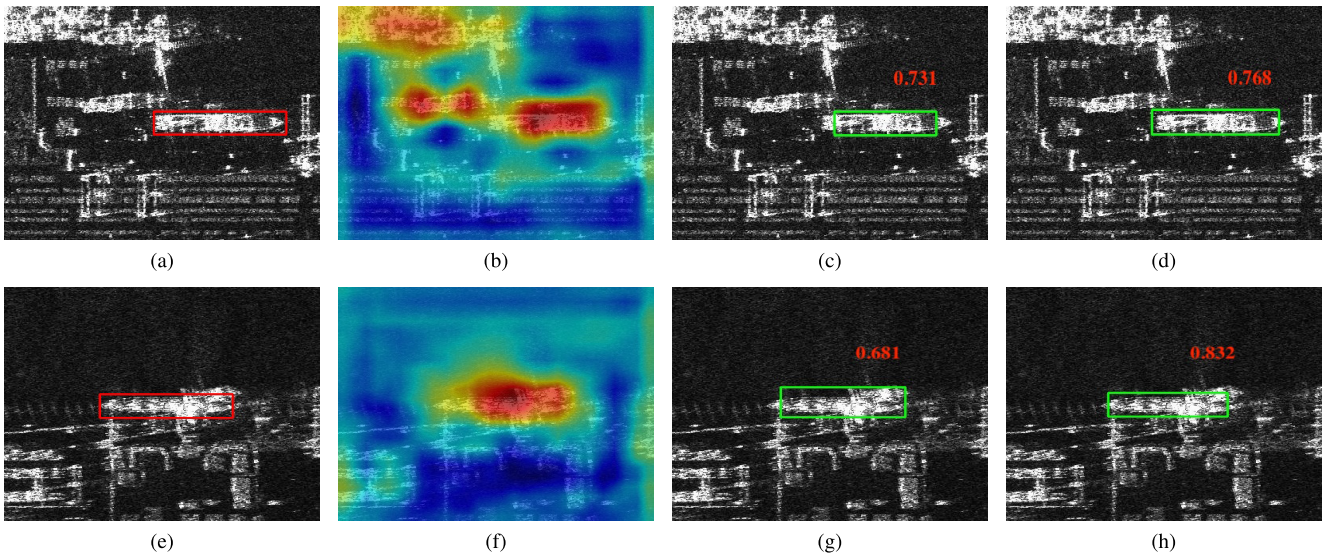


Fig. 6. Some visualization results. The red numbers in the images represent the calculated IoU between the predicted bounding box and the corresponding ground truth. (a) and (e) Ground truth. (b) and (f) Visualization of the confidence maps. (c) and (g) Detection results without the FR module. (d) and (h) Detection results with the FR module.

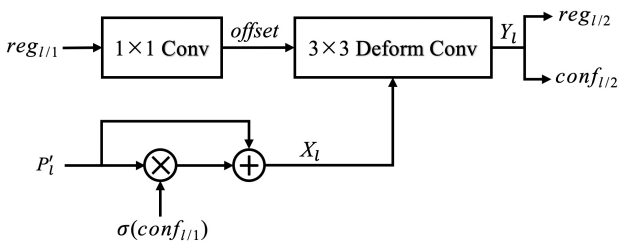


Fig. 7. Structure of the FR module.

The outputs of the first part are denoted as $\text{reg}_{l/1}, \text{conf}_{l/1} \in \mathbb{R}^{(W/s_l)\times(H/s_l)\times C}$. During the training, the same regression and classification loss as in the final prediction part are introduced for the first part and jointly optimized with (11). The structure of our FR module is illustrated in Fig. 7. First, the sigmoid function is applied on $\text{conf}_{l/1}$, whose result is used to generate the adjusted features $X_l \in \mathbb{R}^{(W/s_l)\times(H/s_l)\times C}$ with P'_l as follows:

$$X_l = P'_l \otimes \sigma(\text{conf}_{l/1}) + P'_l. \tag{8}$$

Then, the FR can be expressed as

$$Y_l = N_{\text{FR}}(X_l, \text{reg}_{l/1}). \tag{9}$$

In this process, we adopt the deformable convolution [58] for N_{FR}. Specifically, a kernel offset field is initially produced from $\text{reg}_{l/1}$ by using a 1×1 convolutional layer. Afterward, the 3×3 deformable convolution layer is applied on X_l with the offsets in order to acquire the refined feature maps $Y_l \in \mathbb{R}^{(W/s_l)\times(H/s_l)\times C}$ for further classification and bounding box regression. For each location p_0 on the output Y_l

$$Y_l(p_0) = \sum_{p_k \in \mathcal{R}} w(p_k) \cdot X_l(p_0 + p_k + \Delta p_k) \tag{10}$$

where $\mathcal{R}$ is a regular grid for sampling over the features X_l, for example, $\mathcal{R} = \{(-1,-1),(-1,0),\ldots,(0,1),(1,1)\}$ represents a 3×3 kernel with the dilation rate of 1. $\Delta p_k (k = 1,\ldots,|\mathcal{R}|)$ is the kernel offset. The final output is the weighted summation of the sampling values. In our FR module, the confidence feature maps can help to highlight the ship locations, and the regression feature maps provide the guidance information for the FR. The FR module is placed after two convolutional layers in the general detection head by default in our experiments.

D. Loss Function

Our training loss function consists of three parts

$$L = \frac{1}{N_{\text{pos}}} \sum_l \sum_{x,y} \Big\{ L_{\text{cls}}\big(c^l_{xy}, \hat{c}^l_{xy}\big) + [\hat{c}^l_{xy} = 1] L_{\text{loc}}\big(\boldsymbol{v}^l_{xy}, \hat{\boldsymbol{v}}^l_{xy}\big) + \big[\hat{c}^l_{xy} = 1\big] L_{\text{IoU}}\big(\text{IoU}^l_{xy}, \widehat{\text{IoU}}^l_{xy}\big) \Big\} \tag{11}$$

where $L_{\text{cls}}, L_{\text{loc}}$, and L_{IoU} denotes the classification loss, localization loss, and IoU-prediction loss, respectively. N_{pos} is the number of positive samples. $[\hat{c}^l_{xy} = 1]$ is the Iverson bracket, which means $[\hat{c}^l_{xy} = 1] = 1$ if $\hat{c}^l_{xy} = 1$ holds and $[\hat{c}^l_{xy} = 1] = 0$ otherwise.

Considering the imbalance between the negative and positive samples, the focal loss [27] is adopted for L_{cls}. It can be formulated as

$$L_{\text{cls}}\big(c^l_{xy}, \hat{c}^l_{xy}\big) = \text{FL}(c_t) = -(1 - c_t)^{\gamma} \log(c_t) \tag{12}$$

where

$$c_t = \begin{cases} c^l_{xy}, & \text{if } \hat{c}^l_{xy} = 1 \\ 1 - c^l_{xy}, & \text{otherwise.} \end{cases} \tag{13}$$

Here, L_{loc} adopts the IoU loss as in [59] after getting the decoded bounding boxes and L_{IoU} adopts the binary cross entropy, since the IoU-prediction score ranges from 0 to 1.

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III. Experiments

A. Data Sets and Settings

We conduct the experiments on two data sets: SSDD [60] and AIR-SARShip-1.0 [61]. The SSDD data set contains multiscale SAR ships in different sensors, polarization modes, image resolutions, and scenes. It has 1160 images in total, where the training set includes 928 images and the test set includes the remaining 232 images. All the images are resized into 448 × 448 pixels as the input of the detector in our experiments. The ablation studies are implemented on this data set to evaluate the effectiveness of the proposed improvements. In addition, we also use the AIR-SARShip-1.0 data set to validate the generalization of our method at the end of Section III. It comprises 31 high-resolution large-scale images collected from the GF-3 satellite. We randomly select 21 images as the training and validation data, and the remaining ten images are used as the test data. Each image is cropped into 512 × 512 pixels with the overlap of 256 pixels for training and testing.

We use the pretrained ResNet50 on the ImageNet to initialize the backbone network. All the convolutional layers in the feature pyramid and the detection head follow the same initialization method in [27]. Our model is trained with stochastic gradient descent (SGD) for 60 epochs with a total of eight images per minibatch. The initial learning rate is set as 0.001, which is then divided by a factor of 10 at the 40th and 50th epochs, respectively. We use the weight decay of 0.0001 and the momentum of 0.9. All the experiments are implemented in the PyTorch 1.2 framework and carried out over an NVIDIA RTX 2080 Ti GPU.

B. Evaluation Metrics

We adopt the precision, recall, F1 score, and average precision (AP) to evaluate the detection performance. Precision and recall are two widely used evaluation metrics, which are defined as

$$\text{Precision} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FP}}} \tag{14}$$

$$\text{Recall} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FN}}} \tag{15}$$

where N_{TP}, N_{FP}, and N_{FN} denote the number of true positives (TP), false positives (FP), and false negatives (FN), respectively. TP represents the correctly detected ships. FP indicates the false alarms and FN is the missing ships. In general, a detected bounding box is considered as a true positive when its IoU with the ground truth is higher than 0.5. Otherwise, it is regarded as a false positive. If several bounding boxes are matched to the same ground truth, the result with the highest detection confidence is selected as the true positive. The F1 score is the harmonic mean of the precision and recall rate. It can be calculated as

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{16}$$

The AP metric is used to evaluate the comprehensive quality of a detector. All the output results are first ranked in the descending order of the detection confidence; then, the precision–recall curve (PR curve) is plotted by calculating the precision–recall pair at the different confidence threshold. The AP reflects the shape of the PR curve, which is defined as the mean precision under a set of equally spaced recall rates $S = \{0, 0.01, \ldots, 1\}$

$$\text{AP} = \frac{1}{101} \sum_{r \in \text{S}} \text{Precision}|_{\text{Recall}=r}. \tag{17}$$

As the AP is generally calculated when the IoU threshold being 0.5, we denote it as $\text{AP}_{0.5}$. In this article, we also use the AP metric at a stricter IoU threshold of 0.75 denoted as $\text{AP}_{0.75}$, which can better evaluate the quality of the predicted bounding boxes and reflect the localization accuracy.

TABLE I

Performance of the RetinaNet With Different Anchor Densities and Our Baseline on SSDD

Method	Anchor setting		$\text{AP}_{0.5}$
	#scale	#ratio	
Anchor-based RetinaNet	2	3	0.8726
	3	3	0.8891
	4	3	0.8674
	3	5	0.8602
Anchor-free baseline	n/a	n/a	**0.9161**

C. Model Analysis

1) Anchor Impact and Strong Anchor-Free Baseline: Considering the multiple scales and the orientations of the ships, a large set of anchor boxes with different scales and aspect ratios is always initially designed at each spatial location in the anchor-based methods. It is generally expected that detectors can reach a better performance using a denser sampling method for the anchor boxes. To explore the effect of anchor density, we change the number of scale and ratio of the anchors in a prevalent anchor-based method, RetinaNet [27]. Table I shows that the detection results are sensitive to the settings, and increasing the density cannot gain further improvements, especially when the anchor boxes are oversampled. Thus, it is of great significance to choose an appropriate anchor setting, because a poor choice may hamper the final performance. In contrast, the anchor-free baseline we adopt enjoys several advantages. First, it avoids all the processes related to the anchor boxes by using a simple prediction manner (illustrated in Section II-A) and can achieve a better result (shown in Table I). Second, since there is no need to design the fixed anchor boxes, the detector is more flexible when applied to a new data set. Third, the output variables are much fewer, because the network predicts only one localization result at each pixel point on different pyramid levels, while the anchor-based counterparts need multiple anchor boxes for each location. In this way, a large number of false candidates can be reduced in the scenes with sparse SAR objects, making the optimization more efficient and straightforward. With these advantages, the anchor-free method holds much promise for SAR object detection.

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TABLE II
INFLUENCE OF EACH COMPONENT IN OUR FBR-NET ON SSDD

ABP	Refined Head	Precision	Recall	F1	$AP_{0.5}$	$AP_{0.75}$	Params(M)	Time(ms)
×	×	0.8926	0.9026	0.8976	0.9161	0.5244	31.3	**37.0**
✓	×	0.9142	0.9176	0.9159	0.9359	0.5603	31.3	37.8
×	✓	0.9276	0.9120	0.9197	0.9338	0.5807	32.5	39.2
✓	✓	**0.9279**	**0.9401**	**0.9340**	**0.9410**	**0.5906**	32.5	40.1

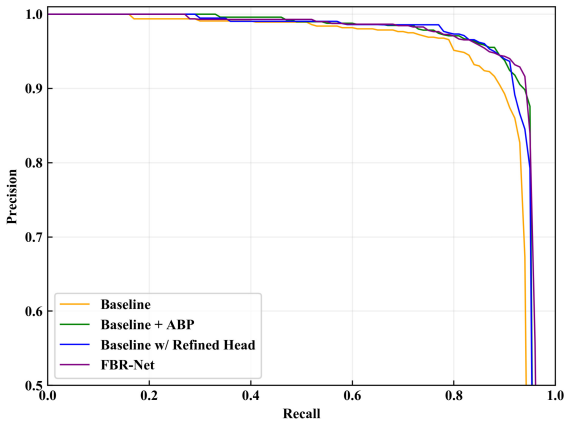


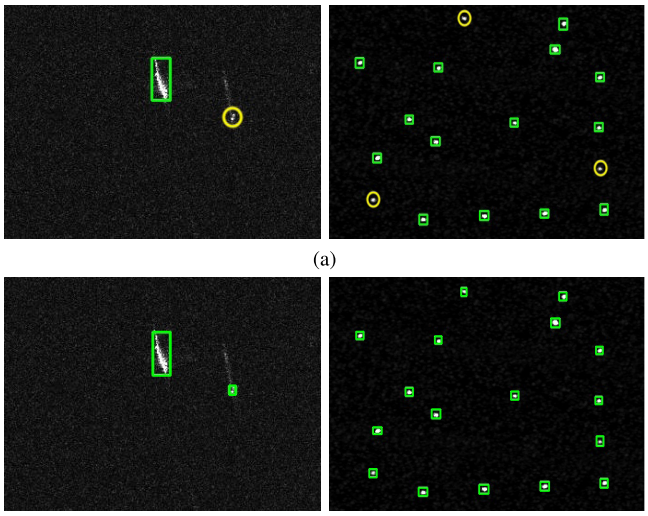
Fig. 8. PR curves of different improvements.

TABLE III
ABLATION STUDIES OF ABP AND THE COMPARISON WITH BFP

ABP		BFP	Precision	Recall	F1	$AP_{0.5}$	$AP_{0.75}$
LA	SA						
×	×	×	0.8926	0.9026	0.8976	0.9161	0.5244
×	×	✓	0.9149	0.9064	0.9106	0.9205	0.5476
✓	×	×	**0.9154**	0.9120	0.9137	0.9311	0.5441
✓	✓	×	0.9142	**0.9176**	**0.9159**	**0.9359**	**0.5603**

2) Ablation Experiments: Two kinds of components are proposed to make the general anchor-free detector more suitable for the multiscale ship detection in the SAR images. In order to analyze the influence of each designed component, a series of ablation experiments is implemented. For a fair comparison, we use the same settings for all the experiments. The overall quantitative comparisons are reported in Table II, and the corresponding PR curves are presented in Fig. 8. It can be seen that both proposed structures can improve the detection performance in different degrees with only a slight increase in the model parameters and the running time per image. Compared with the baseline, the final FBR-Net achieves 2.5% higher $AP_{0.5}$. Furthermore, when the requirement of localization accuracy gets higher and the IoU threshold turns to be 0.75, the $AP_{0.75}$ gains a larger improvement of 6.6%, which means the predicted bounding boxes are more accurate. In the following parts, the effect of each component will be detailed. As the refined detection head contains two independent improvements, both of them will be analyzed separately.

a) Effect of ABP: We first investigate the impact of two submodules in the ABP structure by successively adding them on the baseline. The results are shown in Table III. It turns out



(a)

(b)

Fig. 9. Comparison results of the methods without and with ABP. The green rectangles are the detection results. The yellow circles represent the missing ships. (a) Results of the baseline. (b) Results of the baseline with ABP.

that the LA module brings an obvious improvement, and the SA module can further boost the final performance, especially the $AP_{0.75}$ index. In comparison with the baseline, the ABP finally increases the $AP_{0.5}$ and $AP_{0.75}$ by 2% and 3.6%, respectively. The effectiveness of ABP can also be verified in Fig. 9. We can observe that several small-scale ships are missed by the baseline detector. In contrast, the method with ABP shows a better performance when dealing with the small ships. Then, a further comparison is also made with the BFP [49]. The results show that our method reaches the same level of precision rate with BFP but has a higher recall rate, and the overall performance is better. This is because ABP obtains the balanced features from different levels adaptively with learnable weights but treats all the levels equally. Consequently, the useful semantic information can be extracted more effectively.

b) Effect of IoU-Prediction Branch: The results in Table IV show that the IoU-prediction branch yields a considerable increase of 4.5% in terms of the $AP_{0.75}$, which means that it can help to achieve a higher localization accuracy. The centerness branch [30] can also increase the $AP_{0.75}$ by a large margin, but it only brings a slight improvement on $AP_{0.5}$. In comparison, our method shows a better overall performance. Both methods aim to refine the prediction results by suppressing some low-quality bounding boxes. The difference is that the predicted IoU is also used

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TABLE IV
EFFECT OF CENTERNESS BRANCH AND IOU-PREDICTION BRANCH

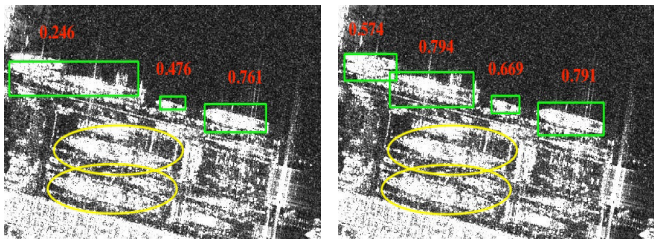
Method	Precision	Recall	F1	$AP_{0.5}$	$AP_{0.75}$
Baseline	0.8926	0.9026	0.8976	0.9161	0.5244
+ Center-ness	0.9047	**0.9064**	0.9055	0.9194	0.5567
+ IoU prediction	**0.9269**	0.9027	**0.9146**	**0.9235**	**0.5697**

TABLE V
RESULTS OF VARYING THE NUMBER OF CONVOLUTIONAL LAYERS BEFORE THE FR MODULE

#conv	$AP_{0.5}$	$AP_{0.75}$
1	0.9240	0.5652
2	**0.9310**	**0.5699**
3	0.9303	0.5576

TABLE VI
EFFECT OF FR MODULE

Method	Precision	Recall	F1	$AP_{0.5}$	$AP_{0.75}$
Baseline	0.8926	0.9026	0.8976	0.9161	0.5244
Baseline + FR	**0.9259**	**0.9120**	**0.9189**	**0.9310**	**0.5699**

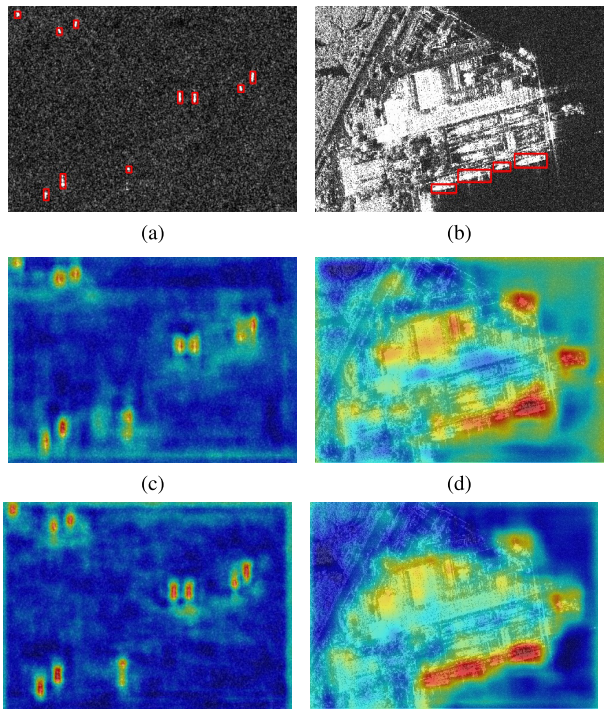


(a) (b)

Fig. 10. Comparison results of the methods without and with FR. The green rectangles are the detection results. The yellow circles represent the missing ships. The red numbers in the images represent the calculated IoU between the predicted bounding box and the corresponding ground truth. (a) Results of the baseline. (b) Results of the baseline with FR.

in our method to guide the final detection boxes, whose effectiveness is proven by the experimental results.

c) Effect of FR Module: We first vary the number of convolutional layers before the FR module, but the total number in the detection head keeps unchanged. Table V shows that the performance of the FR module is insensitive to its position, and placing it after two convolutional layers can achieve a slightly better result. From Table VI, we can see that our FR module offers a significant improvement over the baseline. The 4.6% higher $AP_{0.75}$ manifests its positive effect on improving the quality of the predicted bounding boxes. Furthermore, we also analyze some detection results in Fig. 10. The red numbers in the images represent the calculated IoU between the predicted bounding box and the corresponding ground truth. It can be observed that two ships in Fig. 10(a) are detected as one object by the baseline method. In contrast, the method with the FR module can localize these two ships more accurately, and the generally higher IoU in Fig. 10(b) indicates a more precise result than the baseline.



(a) (b) (c) (d) (e) (f)

Fig. 11. Visualization of the confidence maps. (a) and (b) Ground truth. (c) and (d) Results of the baseline. (e) and (f) Results of the baseline with ABP.

TABLE VII
EFFECT OF OUR METHOD IN TWO DIFFERENT SCENES

Scene	Method	Precision	Recall	F1	$AP_{0.5}$	$AP_{0.75}$
Offshore	Baseline	0.9595	0.9618	0.9607	0.9533	0.5765
	FBR-Net	**0.9713**	**0.9690**	**0.9701**	**0.9644**	**0.6304**
Inshore	Baseline	0.7037	0.6609	0.6816	0.7127	0.3302
	FBR-Net	**0.8273**	**0.7913**	**0.8089**	**0.8006**	**0.4557**

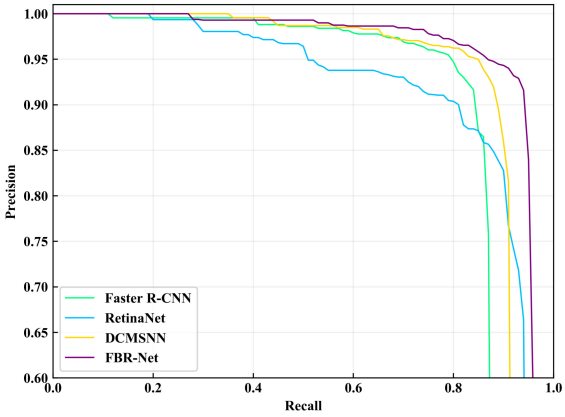


Fig. 12. PR curves of different methods on SSDD.

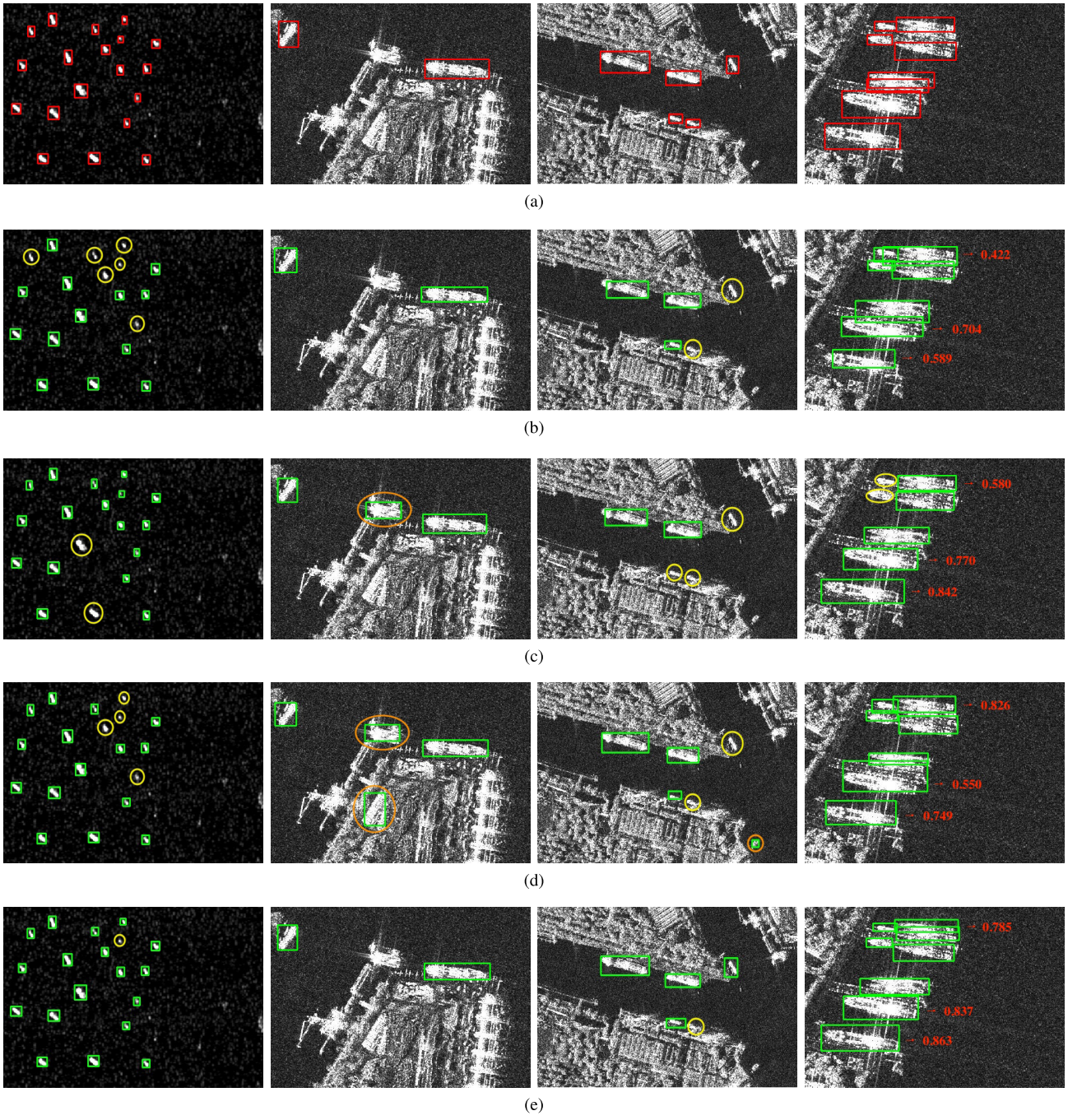
Nevertheless, there is still a problem of the missing ships located in the inland area, which will be discussed in the next part.

So far, we have separately verified the effectiveness of each improvement in the refined detection head. As shown in Table II, the localization accuracy can be further improved by applying these two improvements together.

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TABLE VIII
COMPARISON OF DIFFERENT CNN-BASED METHODS ON SSDD

Method	Anchor-free	Precision	Recall	F1	$AP_{0.5}$	$AP_{0.75}$	Params(M)	Time(ms)
Faster R-CNN	×	0.8728	0.8483	0.8604	0.8528	0.4680	41.1	45.3
RetinaNet	×	0.8310	0.8933	0.8610	0.8891	0.5028	36.1	43.8
DCMSNN	×	0.9049	0.8914	0.8981	0.8943	0.5417	41.1	46.2
FBR-Net	✓	**0.9279**	**0.9401**	**0.9340**	**0.9410**	**0.5906**	32.5	**40.1**



(a) (b) (c) (d) (e)

Fig. 13. Comparison results of four different methods on SSDD. The green rectangles are the detection results. The yellow and orange circles represent the missing ships and false alarms, respectively. The red numbers represent the calculated IoU between the predicted bounding box and the corresponding ground truth. (a) Ground truth. (b) Results of Faster R-CNN. (c) Results of RetinaNet. (d) Results of DCMSNN. (e) Results of our method.

D. Discussion

In this part, we will have some further discussion about our FBR-Net. We first illustrate the effect of two main contributions in our method by the qualitative analysis, respectively. Then, we conduct additional comparison experiments to show the effectiveness of our method in different scenes.

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1) Adaptive Feature Balancing: The proposed ABP structure aims to mitigate the imbalance of semantic information across different levels by balancing the multiscale features adaptively. Table III has validated the effectiveness of ABP. In order to show an intuitive understanding of its effect, some visualized qualitative results are given in Fig. 11. The generated heatmaps are resized to the same size as the input images for a better visualization. As shown in Fig. 11(c), the small ships have a higher score and become more discriminative. This is because the adaptive balanced features produced by ABP can enhance the semantic information of the small ships, which make the detector focus more on these objects and improve the recall rate. It is proven that the adaptive feature balancing plays an important role in the small ship detection. In addition, as for some inshore small ships that are generally difficult to detect, ABP can still pay more attention on them under the strong interference of the harbor facilities, which is shown in Fig. 11(f). Under this circumstance, feature balancing also has a positive influence on reducing the false alarms to a certain degree.

2) Feature Refinement: As mentioned in Section II-C, the feature misalignment is one of the important reasons that limit the localization accuracy in the SAR images. Our FR module aims to refine the object features, so that the enhanced representations enable the features to be well aligned with the ship. As shown in Fig. 6(d) and (h), the FR module improves the IoU between the predicted bounding box and the corresponding ground truth, making the localization more accurate. However, this improvement still cannot address the problem of missing ships located in the inland area, as shown in Fig. 10. The predicted bounding boxes related to these ships are suppressed during the final detection because of the extremely low confidence. We argue that there are two possible reasons for this problem. One is that the quantity of the training samples in this inland situation is too small. The other is that the FR module mainly aims to refine the local features on the basis of the global semantic information of the object, while the inland rectangle bounding box contains too much interference of the background, resulting in the weak semantic representation of the whole ship. In our future work, we will consider using the method based on the oriented bounding box and focus on a better way of combining the imaging mechanism and the object characteristics into the network to achieve a more effective feature representation.

3) Evaluation in Different Scenes: Due to the complex background and the strong interference of the inland area, it is relatively more difficult to detect the inshore ships than the offshore ships. We evaluate the performance of our method in two different scenes. Experiments are carried out in the inshore and offshore scenes, respectively. It can be seen from Table VII that our method brings an obvious improvement in both scenes. Specifically, our FBR-Net increases the $AP_{0.5}$ by 1.1% in the offshore scene, which is mainly attributed to the better detection performance of small ships. As for the inshore ship detection, our method improves the precision rate by a considerable margin, which also means that the false-alarm rate can be significantly reduced. In addition, the much higher $AP_{0.75}$ indicates that our method can achieve more accurate localization.

TABLE IX
COMPARISON OF DIFFERENT CNN-BASED METHODS ON AIR-SARSHIP-1.0

Method	Precision	Recall	F1	$AP_{0.5}$
Faster R-CNN	0.7976	0.7528	0.7746	0.7870
RetinaNet	0.7798	0.7360	0.7572	0.7822
DCMSNN	0.8167	0.7753	0.7954	0.8081
FBR-Net	**0.8944**	**0.8090**	**0.8496**	**0.8458**

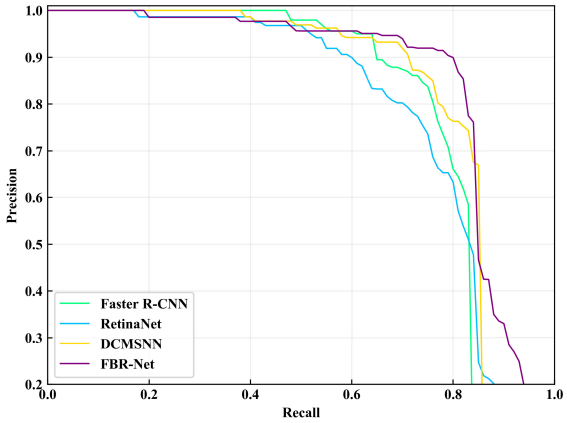


Fig. 14. PR curves of different methods on AIR-SARShip-1.0.

E. Comparison With CNN-Based Methods

We compare the proposed FBR-Net with three CNN-based methods including Faster R-CNN, RetinaNet, and densely connected multiscale neural network (DCMSNN) [39]. The DCMSNN introduces the dense connections to deal with the detection of the multiscale ships in the multiscene SAR images. The detection results of these four methods on the SSDD are listed in Table VIII, and the corresponding PR curves are displayed in Fig. 12. It can be observed that our method improves both the precision and recall rate considerably, achieving the best performance in terms of speed and accuracy. Furthermore, the $AP_{0.75}$ is also increased to a certain degree. It means that we can obtain the predicted boxes with a higher quality, and this is extremely important for some subsequent tasks, such as object recognition. Some detection results are presented in Fig. 13. We can see that the other three methods miss several small ships and generate a few inshore false alarms. In contrast, our method obtains a better performance on multiscale ship detection in the complex scenes, and for some ships, we can get the more accurate bounding boxes. In addition, benefitting from the anchor-free framework, our method is simple and computation-efficient.

In order to verify the robustness and generalization ability of the proposed method, our FBR-Net and other three detectors are experimented on the AIR-SARShip-1.0 data set. Table IX shows that our method still maintains the highest precision and recall rate. The corresponding PR curves are displayed in Fig. 14. Fig. 15 presents the detection results in the large-scale images. It can be observed that DCMSNN misses an obvious

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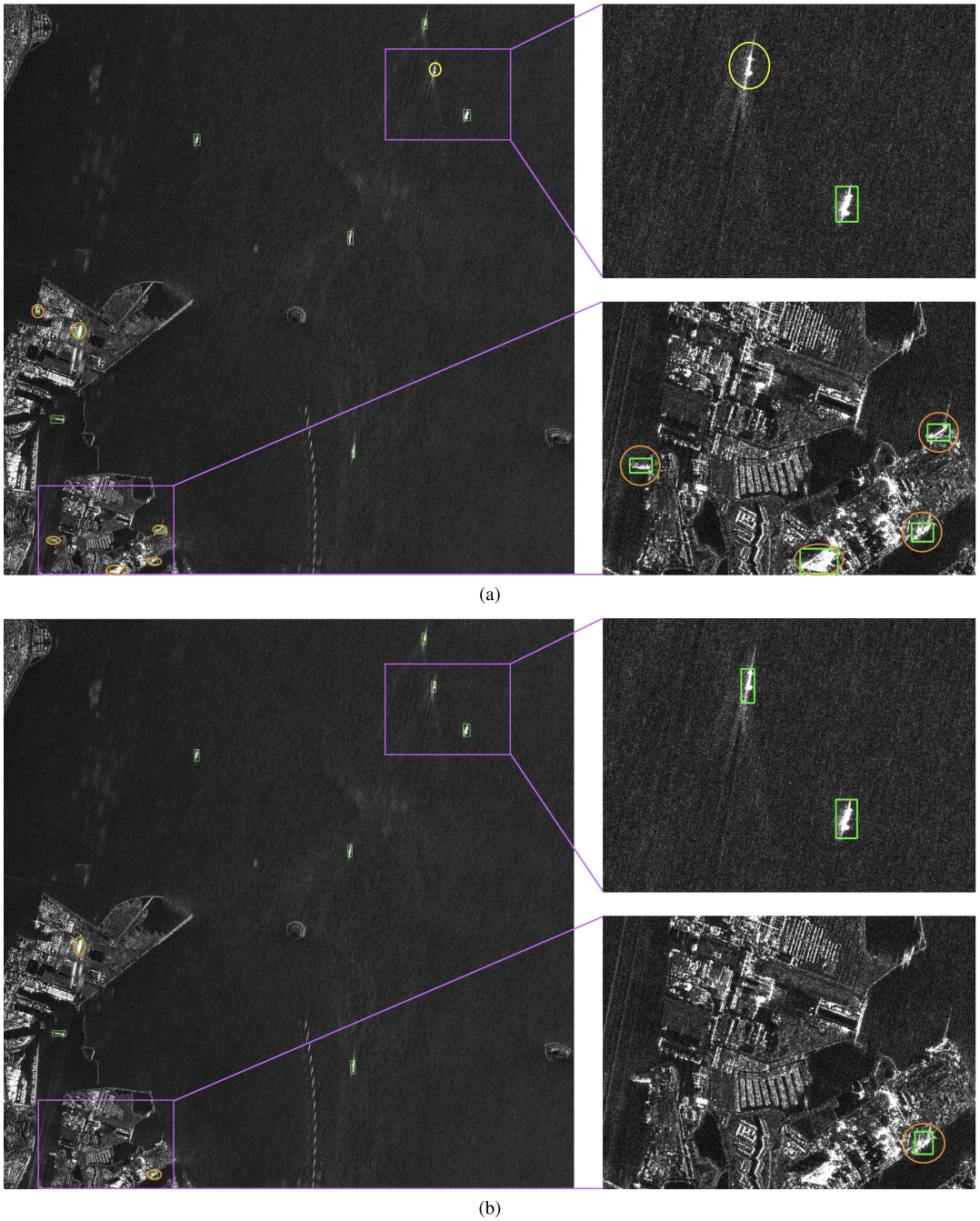


Fig. 15. Comparison results on AIR-SARShip-1.0. The green rectangles are the detection results. The yellow and orange circles represent the missing ships and false alarms, respectively. (a) Results of DCMSNN. (b) Results of our method.

offshore ship and produces more inshore false alarms. By comparison, our method shows a better performance in the large complex scene.

IV. Conclusion

In this article, an anchor-free method based on FBR-Net is proposed for the multiscale SAR ship detection in the complex scenes. We design an ABP to balance semantically the multiscale features across different levels, so that the detector can pay more attention on the small-scale ships. Furthermore, we propose an FR module to deal with the issue of feature misalignment, which is beneficial to the improvement of localization accuracy. The experimental results show that our FBR-Net significantly improves the detection performance over the general anchor-free detector used in the optical images. The comparison with other three prevalent CNN-based detectors also demonstrates the superiority of our method.

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Jiamei Fu (Graduate Student Member, IEEE) received the B.Sc. degree from the Beijing Institute of Technology, Beijing, China, in 2018. She is pursuing the Ph.D. degree with the University of Chinese Academy of Sciences, Beijing, and the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing.

Her research interests include computer vision and synthetic aperture radar (SAR) image processing, especially on SAR object detection and recognition.



Xian Sun (Senior Member, IEEE) received the B.Sc. degree from the Beijing University of Aeronautics and Astronautics, Beijing, China, in 2004, and the M.Sc. and Ph.D. degrees from the Institute of Electronics, Chinese Academy of Sciences, Beijing, in 2009.

He is a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include computer vision, geospatial data mining, and remote sensing image understanding.



Zhirui Wang (Member, IEEE) received the B.Sc. degree from the Harbin Institute of Technology, Harbin, China, in 2013, and the Ph.D. degree from Tsinghua University, Beijing, China, in 2018.

He is an Assistant Researcher with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing. His research interests include synthetic aperture radar (SAR) terrain classification, and SAR target detection and recognition.



Kun Fu (Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees from the National University of Defense Technology, Changsha, China, in 1995, 1999, and 2002, respectively.

He is a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. His research interests include computer vision, remote sensing image understanding, and geospatial data mining and visualization.

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